

Overview

This document presents the newly reorganized directory structure for the Smart Agriculture App. Originally, all codebase components were placed flatly in the root directory. To resolve TypeScript module resolution errors and conform to standard React Native (Expo Router) architecture, the files have been successfully relocated into standard, modular subfolders. All imports now resolve cleanly.

Standard Project Directory Tree

smart-agri-complete/

```
+-- app/                (Expo Router navigation routing folder)
| +-- (tabs)/           (Tab-based main screens & navigation)
| | +-- _layout.tsx     (Tab layout setup & i18n auto-init)
| | +-- index.tsx       (Home / Farmer Dashboard & aggregated stats)
| | +-- insights.tsx    (Alerts & diagnostic recommendations panel)
| | +-- map.tsx         (Interactive map view showing farm pins)
| | +-- profile.tsx     (Settings, language pickers, and support)
| +-- _layout.tsx       (Root layout loader wrapping app context)
+-- components/         (Modular & reusable UI components)
| +-- common/           (Shared components: Card, Loading, EmptyState)
| +-- insights/         (Card views for diagnostic reports)
| +-- maps/             (Interactive Google Maps marker layers)
| +-- settings/         (Configuration popups: Language selector)
+-- hooks/              (Custom React hooks separating logic)
| +-- useFarms.ts       (Farms fetching, sorting, & cache check)
| +-- useInsights.ts    (Diagnostics load, refresh & counts)
| +-- useLocation.ts    (Device GPS coordinate & reverse geocode loader)
+-- i18n/               (Internationalization & translations)
| +-- en.json / hi.json / kn.json (English, Hindi, and Kannada modules)
| +-- i18n.config.ts    (Locale detector config using AsyncStorage)
+-- services/           (Data query APIs & background simulation)
| +-- api.ts            (Central Axios configuration with auth)
| +-- farmService.ts    (Main data requests with offline backup)
| +-- locationService.ts (GPS permission requesting & geocoding)
| +-- mockData.ts       (Simulation data generator fallback)
+-- store/              (Zustand global stores)
| +-- farmStore.ts      (Farms list & detail caching store)
| +-- userStore.ts      (User preferences, themes, & login state)
+-- types/              (Strict TypeScript type definitions)
| +-- farm.ts / insight.ts / user.ts
+-- package.json        (Dependencies: Zustand, Expo Router, i18n)
```